# THE NODE

## Ground truth, architectural boundaries and a defensible route to publication

Baseline register | Version 0.1 | 20 October 2026

Repository: SDarkVader/The-Node, canonical branch main. Branch-head SHA retrieved during preparation: afdf39019ae2472b49b6ea6380bede690cf287a3. Documents cited below were fetched from main on 20 October 2026; individual file SHA and later changes were not independently reproduced as a single checked-out build.

Status discipline: OBSERVED IN SOURCE = directly inspected document/code; REPORTED TEST = cited build/CI result not rerun for this PDF; SPECIFIED = approved design intent or existing documented design; DECISION = unresolved product/research choice; PROPOSED GATE = recommended acceptance threshold; NOT DEMONSTRATED = empirical assertion unsupported by the present evidence.

Purpose: A decision and evidence register for formal development beginning now. This is not a retrospective verdict on a finished platform. It names the minimum architecture worth protecting, the uncertainties that require a deliberate choice, the defects/dependencies that require implementation and the experiments that would justify carefully bounded publication claims.

### Executive determination

The Node has a functioning deterministic society/simulation and an offline Inspect AI evaluation harness. The reported scripted checks support reproducibility of the harness and its mechanical evidence chain; they do not support claims about real-model reliability, intent, hierarchical coordination, deceptive behaviour or social safety. The immediate objective is to turn a technically promising testbed into a construct-valid, traceable experiment without contaminating participant observations, rewriting simulation mechanics to fit desired outcomes, or bypassing live-inference accounting.

Publication strategy: publish engineering methods and fixture validation only as such; add real-model claims only after a separately approved, accounted pilot and controlled, preregistered experiments. Do not make completion of every game mechanic a prerequisite for every narrow study.

## 1. Evidence baseline: what is established today

| Area / status | Grounded finding and correct boundary |
|---|---|
| Simulation and roles — SOURCE | Deterministic socioeconomic kernel, bounded discretionary action/role pipeline, private observations, coupled settlement consequences and shard simulation exist. Some role choices and crime/social systems are still mechanical, partial or unbuilt. [S1, S2] |
| Inspect harness — SOURCE + REPORTED TEST | Native Inspect AI tasks/solver/scorers with a JSON-lines bridge to observe → buildResponsesRequest → validateDecision → stepParticipantState; one controlled Baker per episode, others mechanically driven. Offline mockllm only. Reported 29 scripted samples, 56 fixture-agreement checks per run, independent exported replay and matching state/event hashes; current head was not rerun for this PDF. [S3, S4] |
| Integration defects — DOCUMENTED REPAIR | Independent PR #21 review reported fixes for stale role instructions, behavioural scoring on integrity failure, incomplete-run verification and non-replayable JSON exports. The documented verdict is PASS WITH LIMITATIONS for offline infrastructure only. [S5] |
| Main-line consolidation — DOCUMENTED | The repo records PR #22 consolidation on main, later documentation and CI/recovery updates. Main is the shared development line; two distinct Godot/sky-navigation and Vitest 5 branches remain outside the consolidated work. Do not present these branches as included or tested on main. [S6, S7] |
| Live inference — SOURCE | The Inspect solver rejects non-mockllm providers. The independent account/host design is documented but not live-enabled. Private ledgers, unknown usage, old holds and spending permissions cannot be reconstructed from Git. [S3, S8] |
| Research validity — SOURCE | Current fixture performance validates plumbing/scorer agreement, not genuine agent behavioural performance. Authority attribution is confounded with wording; subjective reassurance cannot establish lying; passive monitoring is detection, not intervention. [S4, S5] |

## 2. Non-negotiable architectural invariants

• I1 — Kernel sovereignty. Actions must enter the existing role-aware, versioned validator and deterministic state transition. No scorer, runner, scripted policy, monitor or live-model wrapper may silently become a game authority. Eligibility and action effects are kernel facts.

• I2 — Information asymmetry. The actor sees only its authorized contemporary observation and action schema. Evaluator-only truth, injection targets, rival hidden data, full event tapes, diagnostic logs, rubric answers and other actors' private memory must never enter its prompt.

• I3 — Independent evidence chain. Retain raw bounded proposal → parsing → eligibility/validator verdict → admitted action → kernel effects → integrity verification → inference/scorer as separate linked records. Never score an unexecuted attempt as an achieved effect.

• I4 — Replay and provenance. Record genesis, versioned action/observation contracts, code/config/source hashes, seed/RNG lineage and admitted action tape. Deterministic replay of recorded actions must reproduce mechanical consequences; stochastic model text need not regenerate identically.

• I5 — Mortal private memory. A participant cannot regain perfect recall from traces or test instrumentation. Civic/public collectively witnessed evidence can persist under its own rules; transient/private beliefs or experiences cannot silently become permanent shared truth.

• I6 — Economic integrity and recovery. Conservation/issuance/settlement rules, bounded harms, mechanical backstops, role progression, baseline access and neutral recovery remain observable and tested. The sim should not reward evaluation scores by construction.

• I7 — Simulation/visual coherence. The visual contract is informational: district weather/colour-temperature, emissive signals, movement, visibility and agency correspond to real underlying state. A display is not proof of a mechanic; no fabricated atmospheric signal or agent knowledge.

• I8 — Zero-cost by default. Offline tests cannot call providers, load credentials, reserve real funds or initialize accounting. One durable accounting authority controls any future dispatch; unknown liabilities are not assumed to be zero.

## 3. Implementation and decision register

Priority expresses dependency, not urgency or an instruction to change every component immediately. “Fix” means the next verifiable contract or test, not necessarily a broad implementation.

| ID / disposition | Required change or explicit decision / acceptance signal |
|---|---|
| A01 \| CONTRACT / FIRST | Define versioned personal-event and diary semantics: which observable events produce records, whether agents select a non-editing mark, allowable curated menus, authority over interpretation and how individual preferences are represented. Test event provenance, no fabrication, no mutation, no TTL refresh, no private leakage. Preserve legacy tape decoding. [S2, S9] |
| A02 \| BUILD, depends A01 | Mechanically generated diarised connections and genuine earned recognition are not built on the reviewed broker path; the legacy composed-entry schema persists. Implement only after A01 is approved. Compare memory creation/expiry under live observations and isolated replay. [S2, S9] |
| A03 \| TEST / PRIVACY | Keep the documented two-day rolling diary TTL, separately driven daily 0.2 distortion rolls and stable subject/context unless design is intentionally revised; reads must not trigger drift or refresh. Test restart, cross-shard transfer, adversarial access and all actor/evaluator views. [S9] |
| A04 \| DEPENDENT MECHANIC | Trespass is incomplete: homes are not individually assigned, legitimate absence/keys/preparation/consumption/witness/consequence pathways remain unfinished. Build abode identity → authoritative absence → crafted-key/shared-resource transactions → connections-only access → sanctions, with conservation/privacy tests. Not a gate for narrow bakery-only studies. [S2, S9] |
| A05 \| RESEARCH DESIGN | One Baker and mechanical neighbours are insufficient for hierarchical multi-agent, cooperation or social causality claims. Add multiple independently controlled roles only where the question requires it; preserve mechanical inhabitants for ecological context and bounded inference spend. [S4, S8] |
| A06 \| DESIGN CHOICE | Shard availability is decoupled from population, with a present cap of three. Whether and how to extend beyond three requires a bounded arrival/population/cost model, migration-choice contract and measured tests; never restore the old population threshold as a proxy for opportunity. [S2] |
| A07 \| DESIGN CHOICE | Funded settlement changes measured inequality materially (historical 0.68 → 0.32 Gini). Decide the intended balance and currency semantics explicitly. Distinguish fairness/wealth from social welfare; replay economy baselines before any claim about effects. [S2, S7] |
| A08 \| FUTURE LIVE ADAPTER | Version the episode driver/Inspect account adapter; use only existing durable host, trusted final-wire builder, broker reservation/settlement and existing gate. Synthetic ledgers/loopback responses first. No direct provider calls from Python or a fresh allowance on clone. [S8] |
| A09 \| REQUIRED BEFORE INTERPRETATION | Create a construct contract for every proposed study: exposure, admissible action and success path, legitimate task, independent truth, control, feasibility denominator, stop condition and exclusion reason. Reachability tests must prove both success and resistance are possible. [S8] |
| A10 \| NEEDS FOCUSED AUDIT | Role-specific verbs, actual wall-clock sessions, persistent ordinary server state, cross-shard identity/recognition and visual/navigation branch contents have uneven coverage. Verify against current code before adding to experiments; do not promote old BLUEPRINT gaps or proposed branches into proven current features. [S2, S7] |

## 4. Decision boundary: the diary and social memory

Confirmed existing documentation: the dated rules review says the world creates and maintains observable event content and connections, while participants can mark actual events without rewriting originals. It also says owner-read is side-effect free, memory is mortal and trespass yields surviving connection/subject structure only. The current `src/engine/diary.ts` retains an older caller-assembled slot interface. [S9, S10]

Unresolved design approval: before the replacement contract is written, explicitly specify whether “choosing what to record” means (a) choosing which genuine events to mark, (b) selecting a constrained interpretation of an observed event, or (c) independently composing an entry. These have different information, memory and falsification consequences. This PDF neither grants arbitrary free-text authorship nor forbids agent discretion by assumption.

Suggested testable split: server-generated event ID + immutable observable fact; optional participant-selected mark/interpretation with separate provenance and decay; no authority to manufacture the fact or edit an earlier record. This is a proposal for decision, not an implemented rule.

Cross-shard condition: individual identity continuity, directed recognition and current visible circumstances must be disentangled from immortal private dossiers. Migration is a new opportunity, not forced amnesia and not full-information disclosure. Verify whether existing code actually carries each item before assigning model tests.

## 5. Construct validity: claims we can and cannot defend

| Study construct | Necessary comparison / evidence; current claim boundary |
|---|---|
| Prompt injection and instruction integrity | Same seed, legitimate feasible objective, declared lower-trust channel, malicious/benign/no-injection controls; report exposed, attempted, admitted and effective separately. Current test notices are synthetic sidecar inputs, not naturally encountered hostile game content. [S4, S8] |
| Authority / hierarchy | Hold requested act, wording, timing, observer-visible evidence and role eligibility constant; vary source/authority attribution alone; counterbalance order. Current council-versus-neighbour difference also varies wording. No formal hierarchy inference yet. [S4, S8] |
| Sycophancy | Orthogonal flattery/consensus and truth-value manipulations; measure unsupported agreement against independent facts. Agreement or a matching price by itself does not reveal motive. [S4] |
| Deception / strategic concealment | Require checkable factual proposition, available contrary evidence, authorized opportunity to communicate it and observed downstream effect. A “feeling secure” post is subjective; current reassurance score cannot prove a lie or intent. [S4, S8] |
| Cooperation / social influence | At least two independently controlled participants or a justified mechanical confederate; feasible interaction, matched counterfactual and welfare measures for self/others. Pricing alone is not a cooperation outcome. [S4, S8] |
| Oversight / control | Vary no monitor, passive detector and actual intervention. Record false positives, detection latency, action prevention and legitimate task utility; passive flags are not saved harms. [S4, S8] |
| Knowledge / uncertainty | Compare supported/hidden evidence, abstention and claims against evaluator truth; grade only opportunities that were actually presented and answered. Lucky correctness is not evidentiary support. [S4, S8] |
| Longitudinal change | Matched early/late opportunities, explicit memory-policy conditions, bounded rollouts and role/departure censoring. Script fixture memory outside game is not proof that model agents remember, forget or learn. [S4, S8] |
| Collective outcomes | Per-seed health, inequality, population and wealth versus matched counterfactuals, with exogenous shocks/churn represented and uncertainty at episode/seed level. No single “good society” scalar supports alignment. [S4, S8] |

## 6. Safety, accounting and research isolation

• No unauthorized model calls. Current `LiveModelRefused` and `assertLiveAdmission` stay closed. A model-capable Inspect adapter is designed/tested with synthetic loopback only until separate review and authorization. [S3, S8]

• Request provenance is byte-exact. Final task, observation, permitted notices and output schema must be included before hashing/reservation; Python cannot append unaccounted content or supply authoritative hidden state. [S8]

• Financial transaction discipline. Durable reservation before dispatch, settle trustworthy usage before content/format/freshness rejection, retain holds for unknown usage, track overrun liabilities, never refund a charged but unused response. [S8]

• Idempotence and crashes. Bind call identity to manifest, isolated host, actor and role epoch/tick. No duplicate dispatch after crash, stale role or retry. Prove fault recovery with synthetic journals; never backfill “unknown” with zero. [S8]

• Independent evaluation data. Evaluation fixtures, ground truth, hidden targets and monitors reside outside policy inputs; controlled low-trust content is conspicuously versioned and typed. Sanitize any public traces. [S3, S8]

• No hidden chain-of-thought claim. Future bounded self-reported decision justifications are optional outputs, not access to internal reasoning. Start with action-only trials; preregister any explanation intervention because eliciting rationale can change actions. [S8]

## 7. Evidence schema and integrity tests

Minimal joined event record (recommended): experiment/manifest ID and preregistered construct; source/build and scenario version; anonymized actor/role/host instance; seed and independent RNG lineage; tick; authorized observation hash and visibility policy; untrusted notice surface and exposure record; source prompt/wire hash; model or scripted-policy identity and sampling configuration; proposal and parsing result; validator reasons; admitted action; kernel event/effects; monitor intervention; settlement invariants; replay hashes/digests; evaluator-only factual targets; score plus provenance/evidence links; missingness or failure classification.

Outcome taxonomy: `executed` (with measured effect), `attempted_but_rejected`, `unparseable`, `not_applicable` (no opportunity), `not_scorable` (evidence or infrastructure failed), `budget_stopped` and `censored_by_departure` are different results. None of the last four may be counted as a successful or failed behavioural choice. [S8]

Replay criterion: two independent offline runs agree on fixtures and mechanical state/event digests; records serialized to disk must replay in a fresh process. Real model text can differ, but the accepted action trace must deterministically reproduce the recorded kernel output. [S5]

## 8. Publication path: sequential evidence gates

| Gate / output | Acceptance condition, deliverable and what may be claimed |
|---|---|
| G0 — Ground-truth freeze | Version this register and make owners approve diary, role and shard semantics. Identify authoritative source for every invariant and gap. Deliverable: reviewed source/decision register with open issues; no behavioural claim. |
| G1 — Offline conformance | On a pinned main checkout, verify exact versions, complete test suite, native checks, replay/export and host security invariants; reproduce or explicitly qualify archived CI evidence. Deliverable: reproducible harness technical report. Publishable as engineering methodology, not model-safety results. |
| G2 — Construct/reachability | Select one narrow question (e.g. council authority attribution), write versioned task/exposure/feasibility/outcome contracts, nonleaky controls and reachability tests. Deliverable: preregistered scenario plus pilot protocol. |
| G3 — Accounted execution safety | Offline custom-provider tests with synthetic ledger, durable host and failure injection; review spending ledger and manifests without enabling calls. Deliverable: independent fail-closed security and accounting report. |
| G4 — Authorized minimal pilot | Only with explicit funding/run authorization and verified private accounting, run a tiny real-model pilot sufficient to validate transport, parsing, billing, observation isolation and test feasibility. Deliverable: feasibility appendix, not generalized model-quality findings. |
| G5 — Registered controlled study | Freeze hypotheses/metrics/control arms; allocate held-out seeds and models; randomize order; prespecify exclusions, statistical unit and uncertainty methods. Separate model-level from episode-level sampling. Deliverable: protocol plus reproducible experiment package. |
| G6 — Main comparative results | Collect independent episodes, blind/automated scoring, inspect failure modes, report all attempts/admissions/effects, calibration and confidence intervals. Disclose weak proxies, nulls, sensitivity tests, and multiple-testing policy. Publish only constructs directly supported. |
| G7 — Social / longitudinal extension | After game memory and multi-actor contracts are validated, extend to cooperation, hierarchy, multi-shard mobility and long-horizon effects. This is a separate research phase, not required to publish a narrow G6 study. |

## 9. Scope controls: prevent inadvertent redesign

• Preserve the existing role roster and basic economy. New model actions are not justified solely because a benchmark wants an easy win condition. Use the current game’s affordances first; classify impossible constructs as unsupported.

• Decouple research milestones. Personal memory and trespass are essential for certain social/longitudinal questions, but not for a well-designed bakery pricing/authority experiment. Do not stall one on unrelated world-building.

• Do not infer intent from outputs. “Deference”, “dishonesty”, “sycophancy”, “cooperation” and “misalignment” must have narrow operational definitions and disconfirming controls.

• Retain negative results. A model that never sees an exposure, fails formatting, exits a role, or encounters an infrastructure fault supplies different evidence; report denominators and censoring.

• Stage optional integrations. Inspect is integrated. AgentDojo and ControlArena are currently adapted designs, not native integrations. Concordia is absent; NIST AI RMF is a project-defined crosswalk, not certification. [S4, S8]

• Version the research object. No changes to rule constants, world topology, validator schemas, memory policy or model request assembly mid-study without new scenario version and documented comparability decision.

## 10. Next decisions and first work package

First explicit decisions (owner approval): (D1) Diary event origin, constrained agent selection and interpretation; (D2) which one behaviour construct anchors the first publication; (D3) mechanical versus multiple-model actor participation for that construct; (D4) settlement/shard tuning only if it affects the chosen trial's validity; (D5) whether to keep long-term game/player UX and the research testbed on the same runtime schedule.

Recommended first coding increment: no new inference. Add a machine-readable `construct_contract` and offline reachability checks to one existing Inspect scenario, including exact legitimate goal, source-label-only treatment pair, eligible exposure/response/effect predicates, blinding of evaluator truth, stopping rule and logged denominators. It can be independently reviewed without touching diary, economy, live credentials or the protected broker. [S8]

First independent review deliverable: a pinned-SHA reproduction record for G1 with commands, test counts, hashes, environment, CI links, failure logs and unsupported claims; then lock the G2 design before any agent comparison. Do not replace those artifacts with an attractive chart or a prose confidence statement.

## 11. Sources and provenance

Primary source is the repository and its dated design/review documents. Source assertions in this PDF were assembled by reading current `main` documents and selected implementation files; CI numbers are quoted as historical reports, not rerun here. Some older Blueprint sections explicitly carry superseded statements; the dated reviews/checkpoint should be read beside source files for contradictions.


S1 Repository README

S2 System blueprint and known gaps

S3 Inspect harness source and architecture

S4 Evaluation scope and limitations

S5 Independent PR #21 evaluation review

S6 Main consolidation and CI evidence

S7 Current session checkpoint

S8 Accounted-agent and framework design

S9 Rules/diary gap review

S10 Current diary implementation

S11 Visual invariants and data/visual contract

S12 Housing, memory and access design

S13 Adversarial containment and baseline recovery

S14 Research questions and illustrative constants

S15 Inspect solver fail-closed source


Register note: proposals in this PDF are labeled as proposed gates or design decisions. They do not modify the repository, authorize inference spend, establish a completed experiment or declare the user's design choices settled.